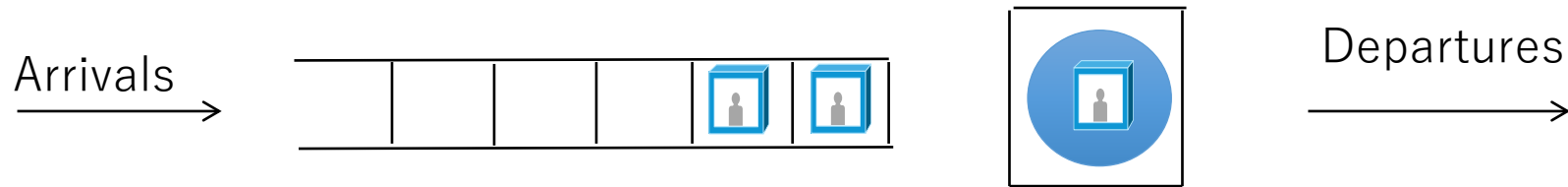


# 数理工学モデル化

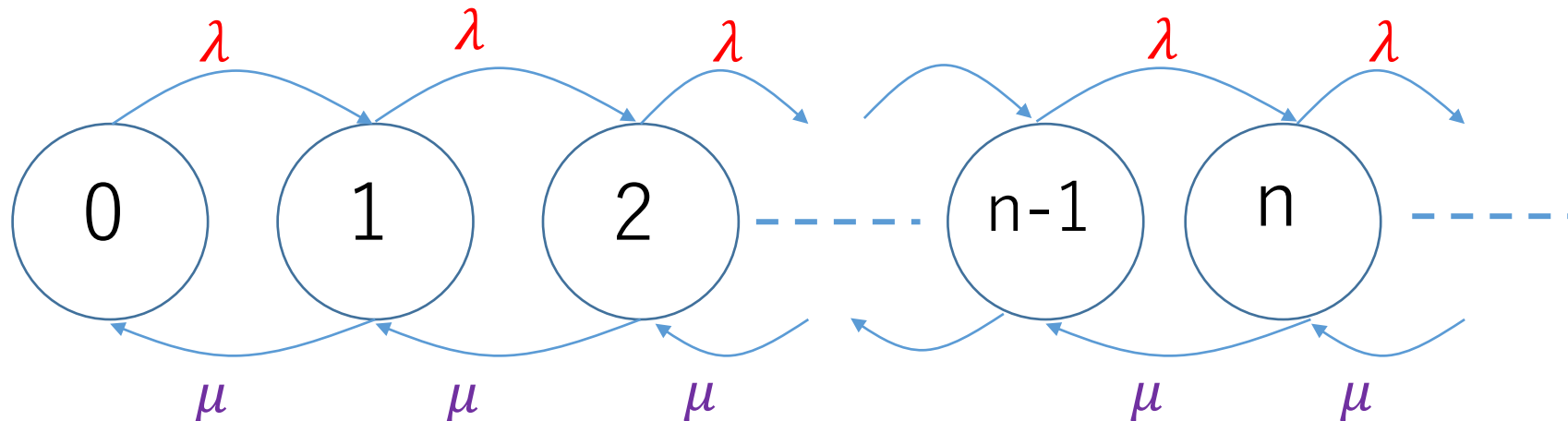
担当：フンドック トゥアン

# Single server model (単一サーバモデル)



# M/M/1 model: Single model

- M/M/1 model: M/M/1待ち行列
  - Exponential arrival interval with mean:  $1/\lambda$  (平均到着間隔)
    - $\lambda$  1 単位時間あたりに平均到着客数
  - Exponential service with mean:  $1/\mu$  ( $\lambda < \mu$ ), (平均サービス時間)
    - $\mu$  単位時間あたりにサービスできる平均客数
  - Traffic intensity:  $\rho = \lambda/\mu$ : トラフィック強度
    - サーバが稼働中の確率
- $N(t)$ : # of customers in the system at time  $t$ : 時刻  $t$  の系内客数
  - $\{N(t); t \geq 0\}$  forms a Markov chain on state space  $\{0, 1, 2, \dots\}$
  - Birth-and-death process (出生死滅過程)



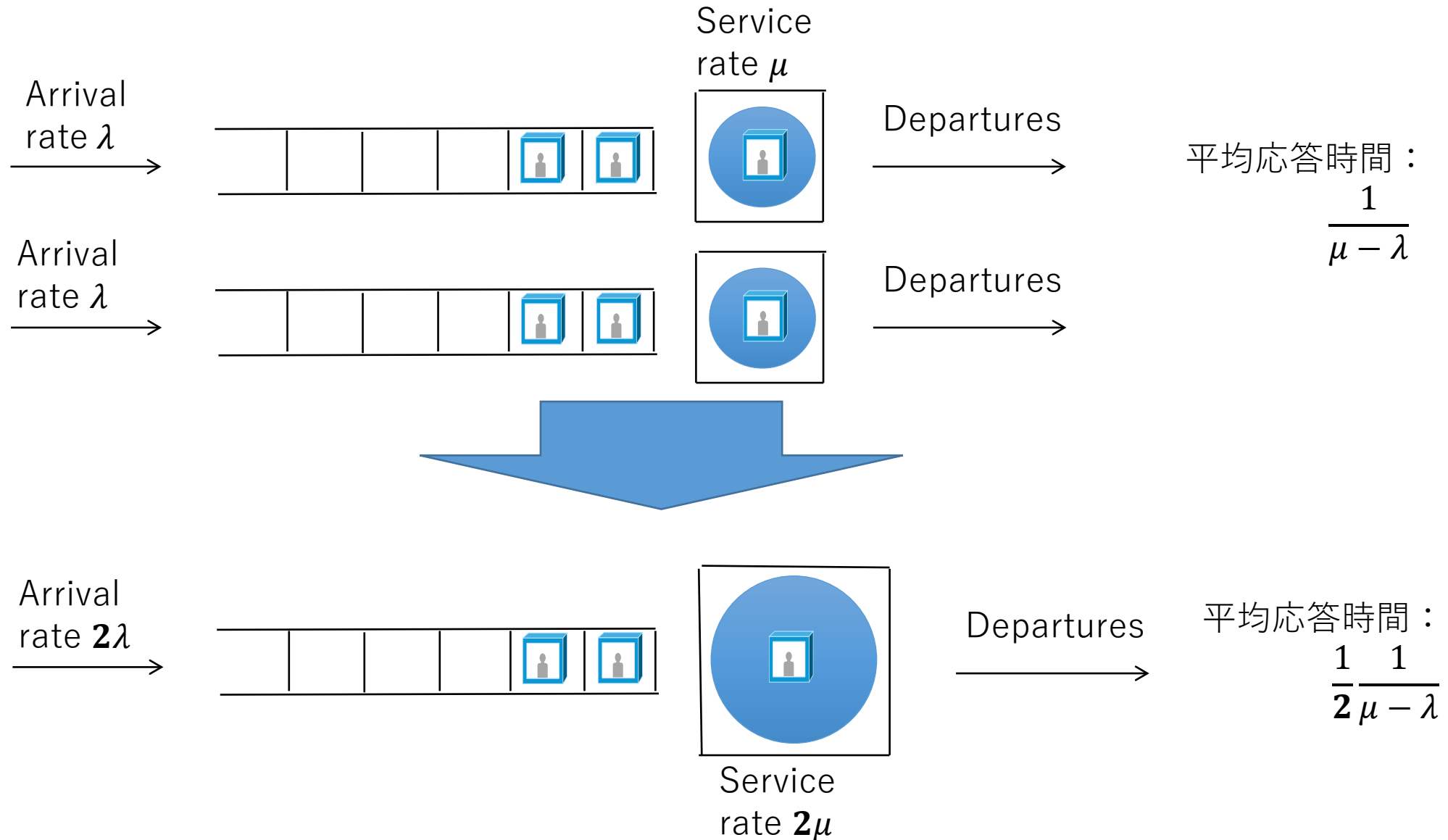
# Results for M/M/1 model :

## M/M/1モデルの結果

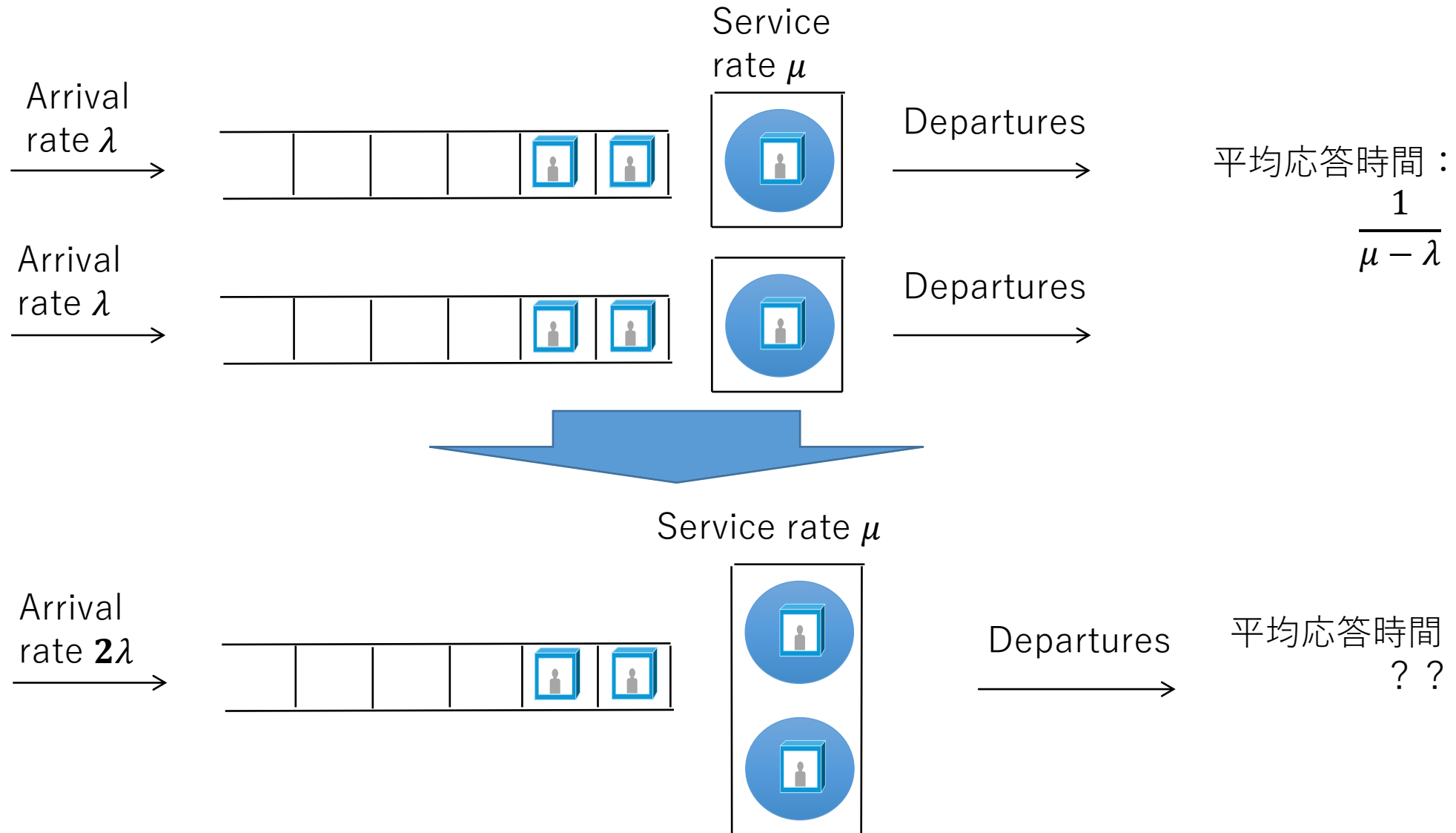
- Stable condition: Queue is not divergent (安定条件：並ぶ客数が無限に発散しないための条件)
  - $\lambda < \mu$
  - $\rho = \lambda/\mu < 1$ .
- Distribution of the # in the system:  $N$  (系内客数の分布)
  - $P(N = n) = (1 - \rho)\rho^n \Rightarrow$  幾何分布
- Mean number of customers in system (平均系内客数)
  - $E[N] = \frac{\rho}{1-\rho}, \rho=0.8 \Rightarrow E[N] = 4$ .
- Mean response time: from arrival to departure (応答時間)
  - $E[T] = \frac{1}{1-\rho} \frac{1}{\mu} = \frac{1}{\mu-\lambda} \Rightarrow \mu = 5, \lambda = 4 \Rightarrow 1, \lambda = 3 \Rightarrow E[T] = \frac{1}{2}$

Economies of scale (規模の経済) :  $\lambda \rightarrow a\lambda, \mu \rightarrow a\mu \Rightarrow E[T^{(a)}] = \frac{E[T]}{a}$ . 応答時間が  $a$  倍早くなる ! ! !  
*Response time is reduced by  $a$  times.  $a=2 \Rightarrow$  half reduced.*

# Economies of Scale (規模の経済)



# Economies of Scale (規模の経済)





Merge two  
small  
convenience stores  
to a bigger  
one?

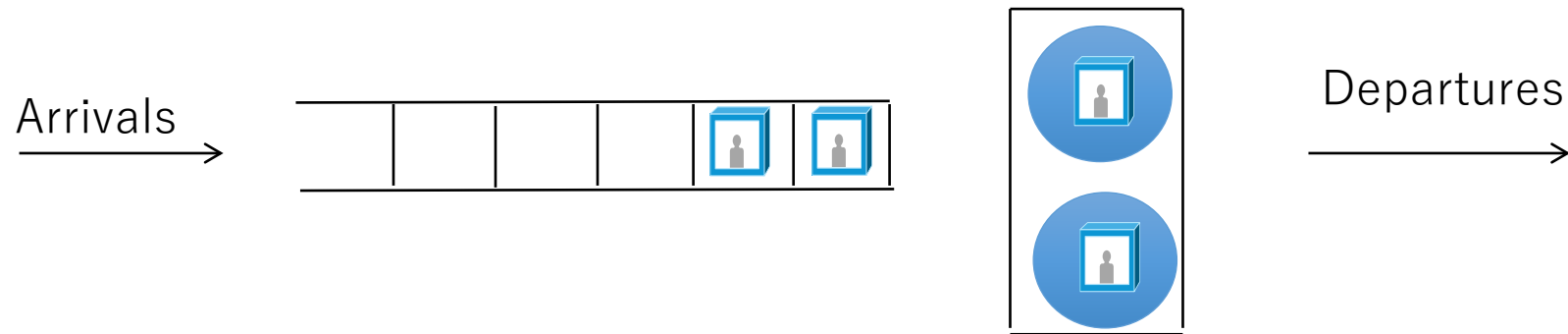


<https://www.itmedia.co.jp/business/articles/1510/29/news039.html>



二つの小さいコンビニ  
は大きな一つのコンビニ  
に合併？

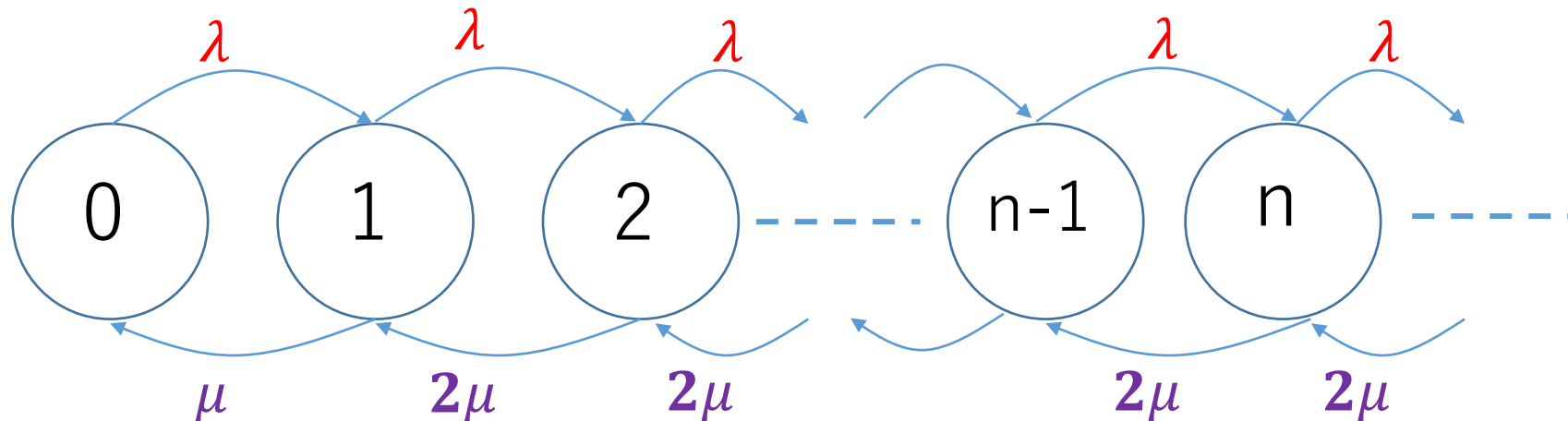
# Multiserver model





# M/M/2 model: two-server model

- M/M/**2** model
  - Exponential arrival interval:  $1/\lambda$  (到着間隔が平均 $1/\lambda$ )
  - Exponential service with mean:  $1/\mu$  ( $\lambda < 2\mu$ ), (サービス時間平均 $1/\mu$ )
  - Traffic intensity:  $\rho = \lambda/\mu$
- $N(t)$ : # of customers in the system at time  $t$  時刻  $t$  の系内客数
  - $\{N(t); t \geq 0\}$  forms a Markov chain on state space  $\{0, 1, 2, \dots\}$
  - Birth-and-death process



# Results for M/M/2 model: two-server model

- Stable condition: Queue does not divergent
- 安定条件（系内客数が無限大に発散しない）
  - $\lambda < 2\mu$ ,  $\rho = \lambda/\mu < 2$ .
- Waiting probability（待つ確率）
  - $$P_W = \frac{\frac{\rho^2}{2}}{\left(1 - \frac{\rho}{2}\right)(1 + \rho) + \frac{\rho^2}{2}} = \frac{\rho^2}{2 + \rho}$$
- Mean number of customers in system（平均系内客数）
  - $$E[N] = \rho \left(1 + \frac{P_W}{2 - \rho}\right) = \frac{4\rho}{4 - \rho^2}$$
- Mean response time: from arrival to departure（平均滞在時間）
  - $$E[T] = \frac{E[N]}{\lambda} = \frac{4}{4 - \rho^2} \frac{1}{\mu}$$

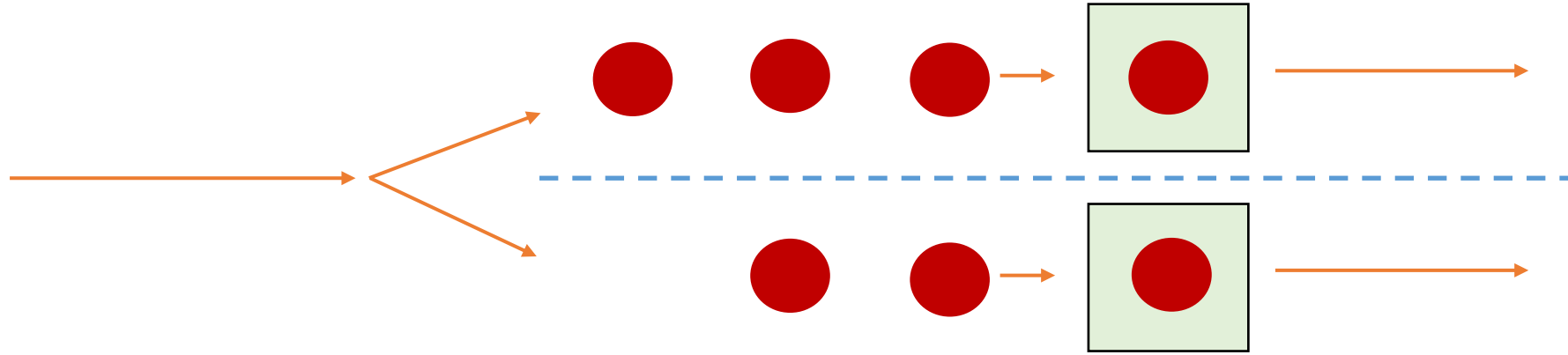


Merge two  
small  
convenience  
stores  
to a bigger  
one?

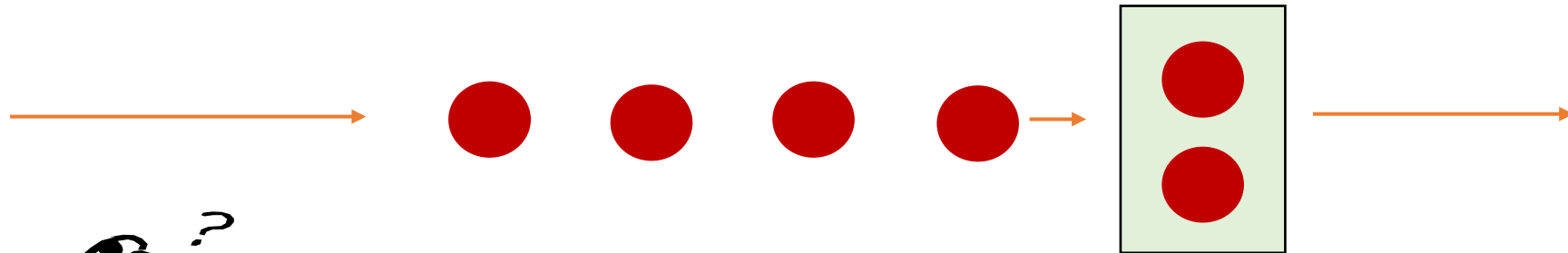


<https://www.itmedia.co.jp/business/articles/1510/29/news039.html>

# Convenience store



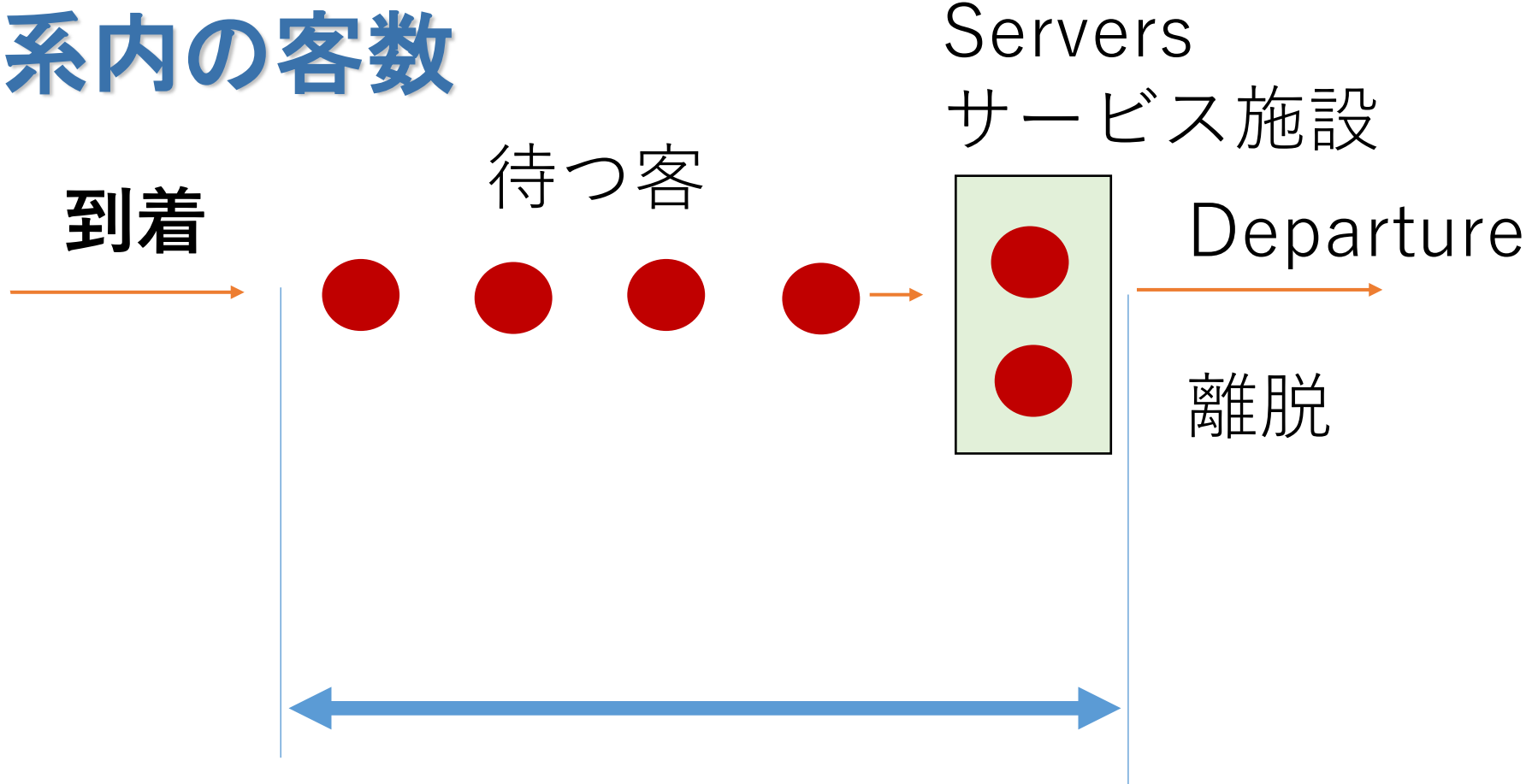
# ATM in a bank



Which system is  
less congested ?

# Congested level=Number of customers

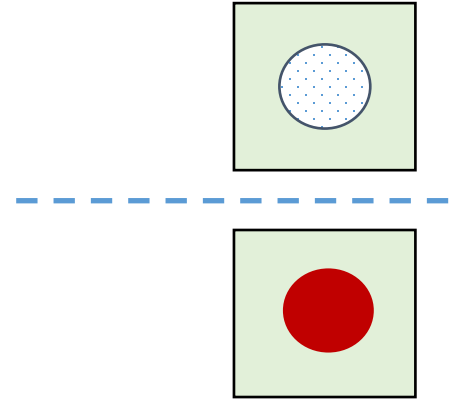
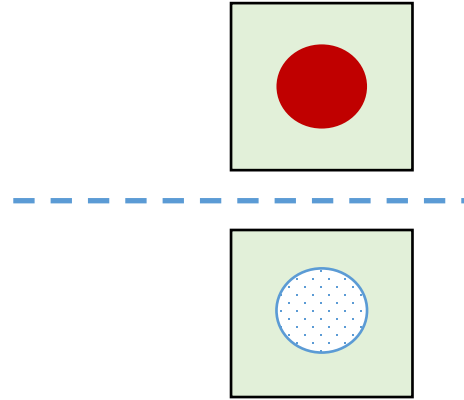
## 混雑度=系内の客数



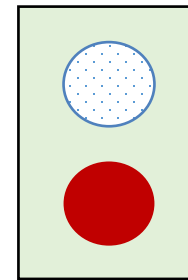
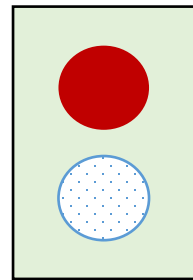
Number of customers (系内客数) =  
Waiting customers (待つ客数) +  
Customers in service (サービス中客数)

Number of customers = 1

Convenience store

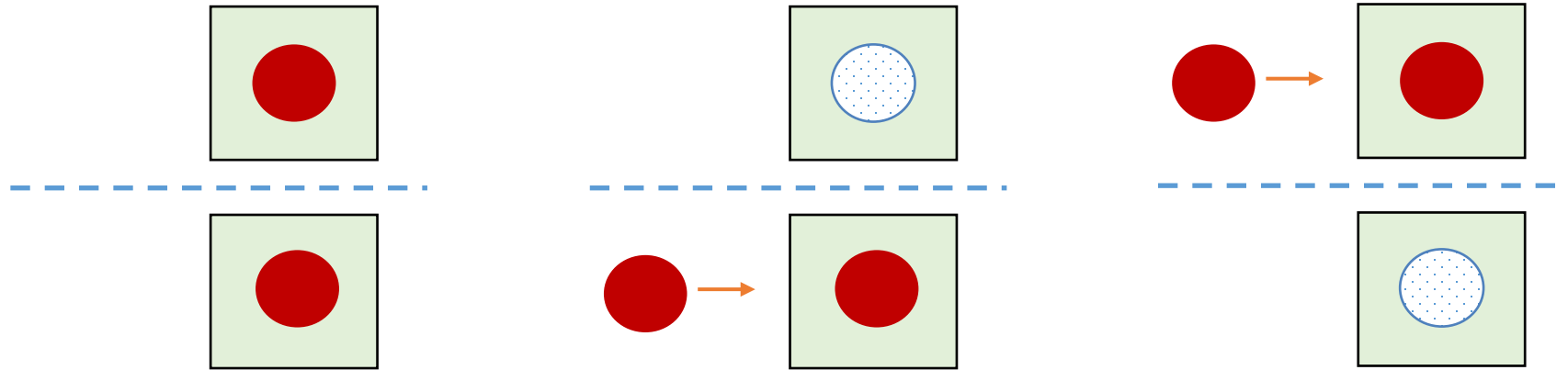


ATM in a bank

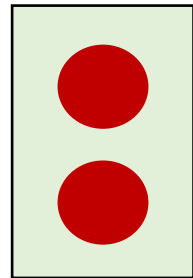


The same

# 客数=2の場合 (Two customers) コンビニ方式 (Convenience store)



## 銀行方式 (Bank policy)

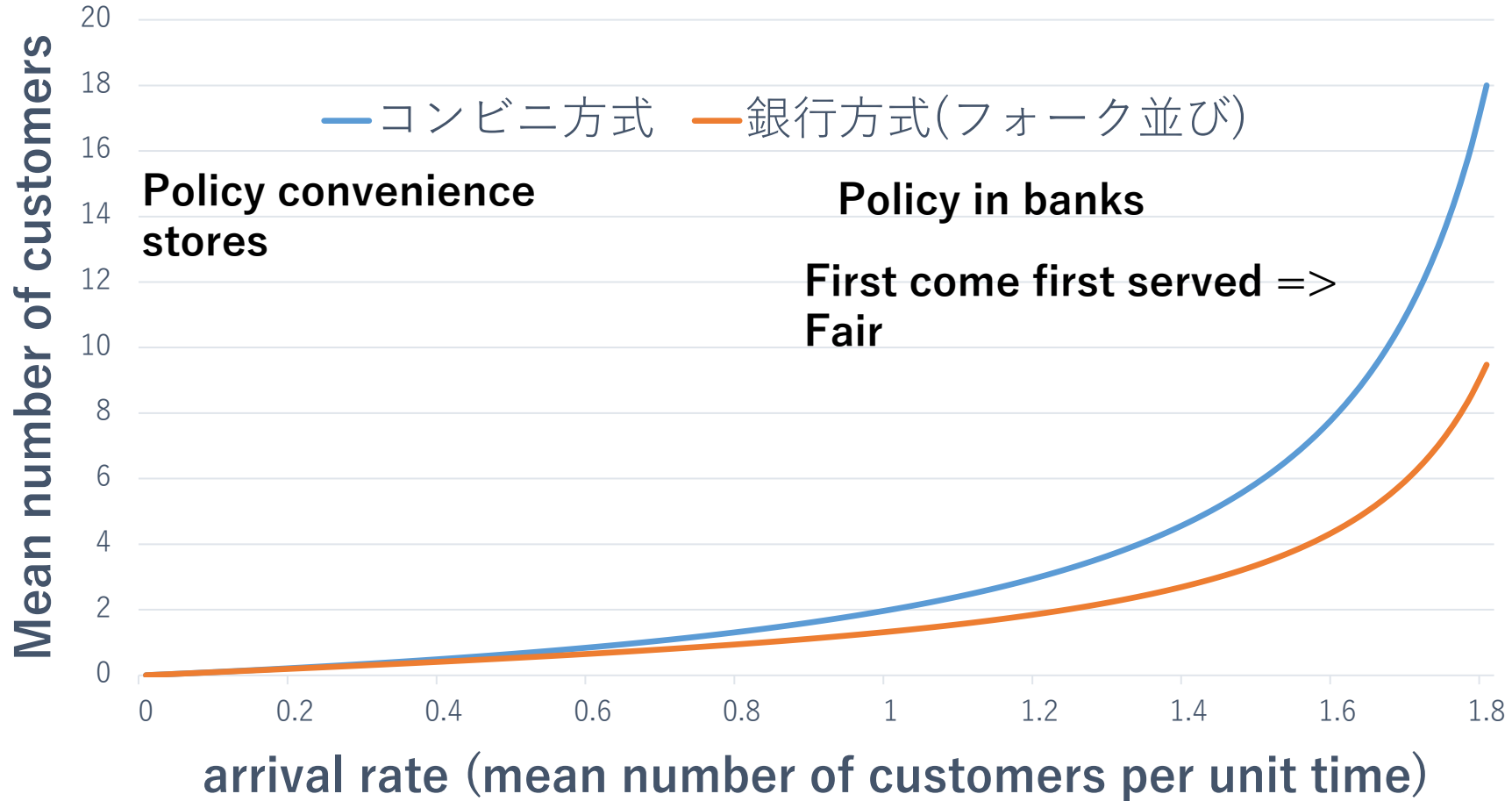


銀行方式は窓口が利用  
されている割合が高  
い？

Bank-policy is more  
efficient (used  
more)?



# Queueing Theory

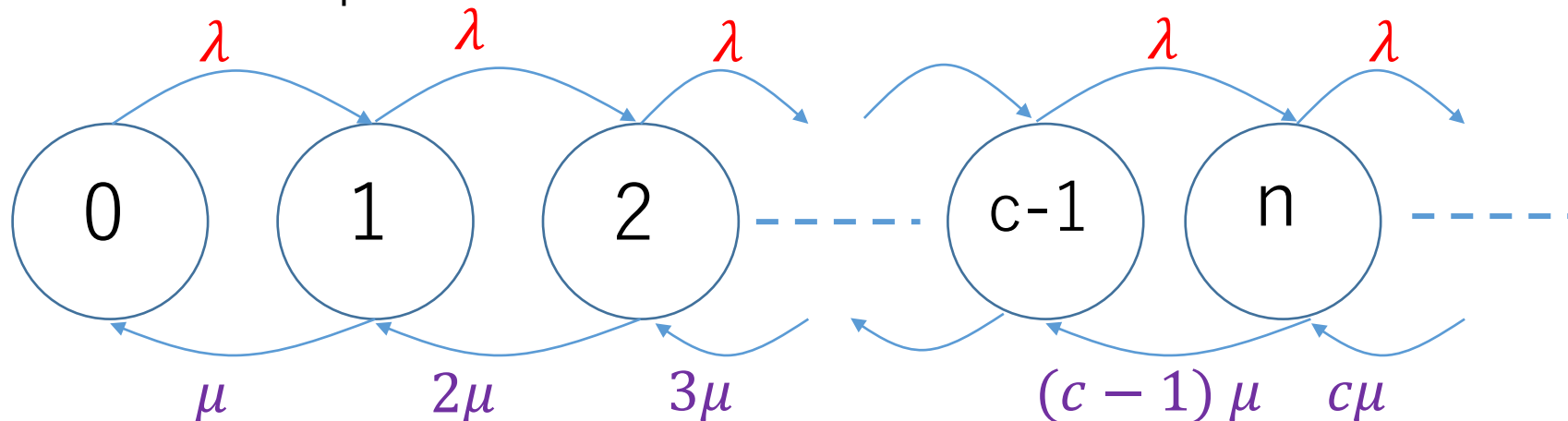


Under some specific assumptions



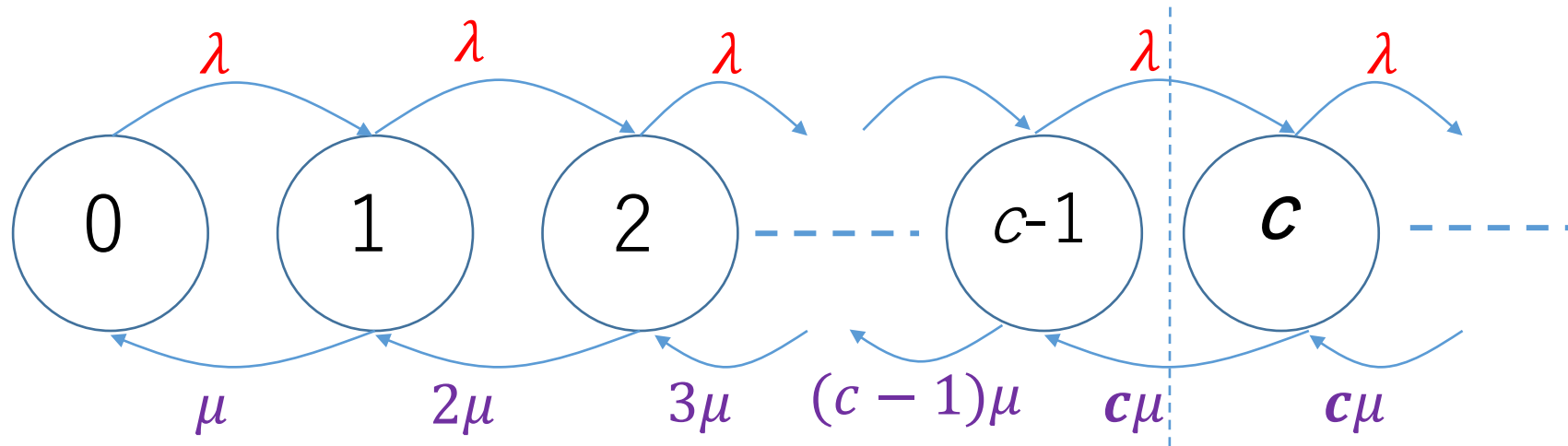
# M/M/ $c$ model: $c$ -server model (複数サーバ)

- M/M/ $c$  model
  - Exponential arrival interval:  $1/\lambda$  (平均到着間隔)
  - Exponential service with mean:  $1/\mu$  ( $\lambda < c\mu$ ), (平均サービス時間)
  - Traffic intensity:  $\rho = \frac{\lambda}{\mu} < c$
- $N(t)$ : # of customers in the system at time  $t$  系内客数
  - $\{N(t); t \geq 0\}$  forms a Markov chain on state space  $\{0, 1, 2, \dots\}$
  - Birth-and-death process



# M/M/ $c$ model: c-server model

- $N(t)$ : # of customers in the system at time  $t$ 
  - $\{N(t); t \geq 0\}$  forms a Markov chain on state space  $\{0, 1, 2, \dots\}$
  - Birth-and-death process
  - $\pi_n = P(N = n), n = 0, 1, 2, \dots$  in steady state (system runs for a long enough time)



$$\lambda\pi_{n-1} = n\mu\pi_n, \quad n = 1, 2, \dots, c \quad \lambda\pi_{n-1} = c\mu\pi_n \rightarrow \pi_n = \frac{\rho}{c}\pi_{n-1}, \quad n = c, c+1, \dots$$

# Results for M/M/ $c$ : M/M/cモデルの結果

- The number of servers is  $c$  サーバ数
- Arrival rate  $\lambda$ 
  - In average  $\lambda$  customers arrive in a time unit
  - The arrival interval between two consecutive customers is  $1/\lambda$
- Service rate  $\mu$ 
  - In average  $\mu$  customers can be served by a server per a time unit
  - Mean service time of a customer is  $1/\mu$

Waiting probability:  $P_W = \frac{\frac{\rho^c}{c!}}{\left(1 - \frac{\rho}{c}\right) \sum_{k=0}^{c-1} \frac{\rho^k}{k!} + \frac{\rho^c}{c!}}, \quad \rho = \lambda/\mu.$

待つ確率

平均系内客数:  $E[N] = \rho \left(1 + \frac{P_W}{c - \rho}\right)$       平均滞在時間:  $E[T] = E[N]/\lambda$

# Queueing Theory (待ち行列理論)

- Average per a unit time (1 単位時間当たり)

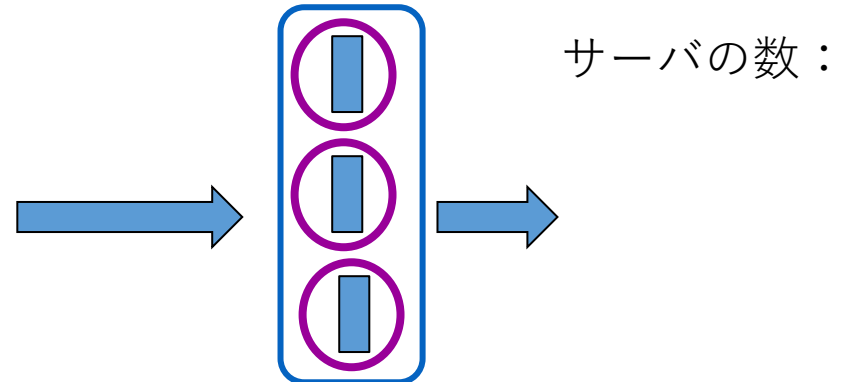
- $\lambda$  customers arrive ( $\lambda$  人到着)
- Each server serves  $\mu$  customers
  - $\mu$  人サービスされる.

$$\rho = \frac{\lambda}{\mu} (\text{Erlang})$$

$$P_B = \frac{\frac{\rho^c}{c!}}{\sum_{i=0}^c \frac{\rho^i}{i!}}$$

## Erlang B

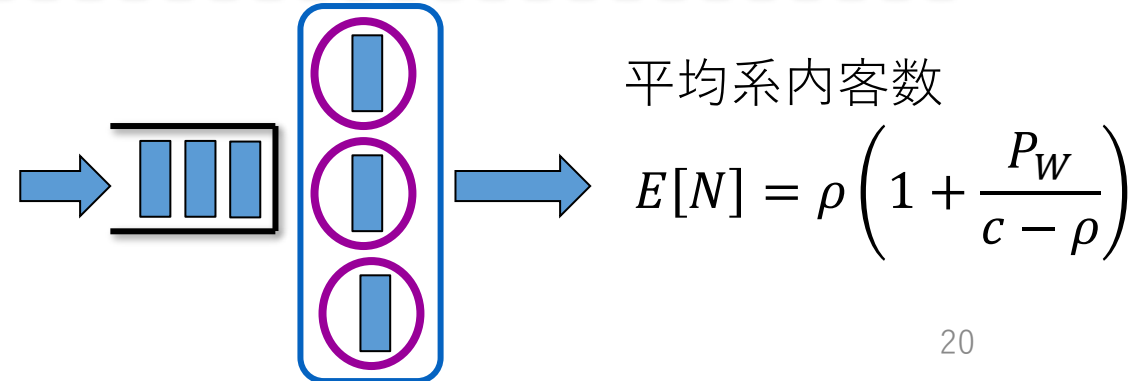
Blocking  
probability  
棄却率



$$P_W = \frac{\frac{\rho^c}{c!} \frac{c}{c - \rho}}{\sum_{i=0}^{c-1} \frac{\rho^i}{i!} + \frac{\rho^c}{c!} \frac{c}{c - \rho}}$$

## Erlang C

Waiting  
probability  
待つ確率

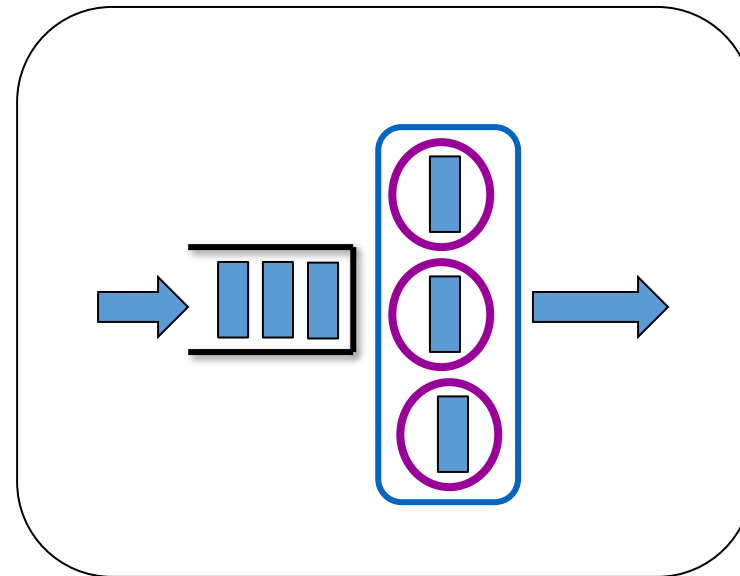


$$E[N] = \rho \left( 1 + \frac{P_W}{c - \rho} \right)$$

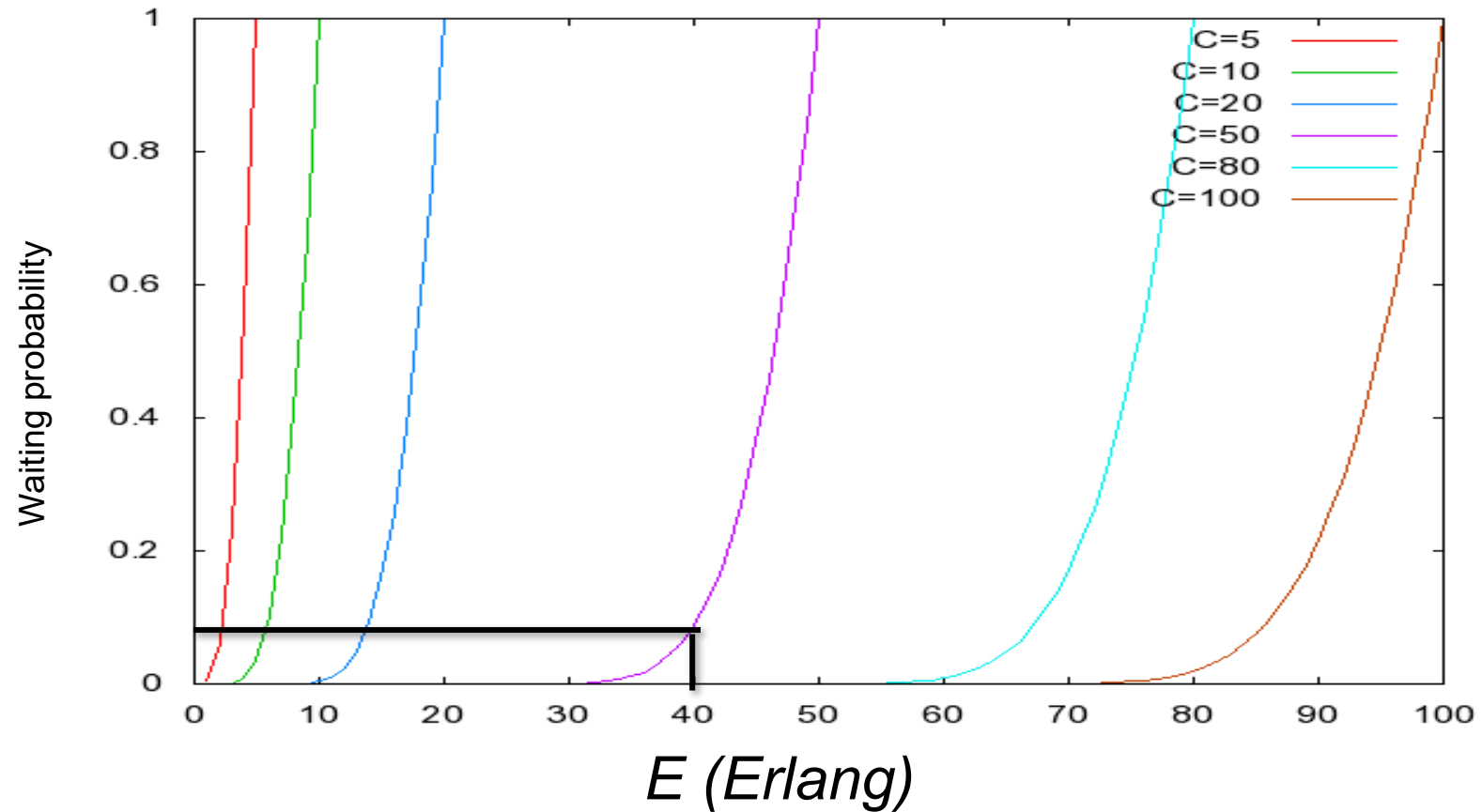
# Application in a call center (コールセンター)

- Model for a call center
  - Operator => Server
  - Message
    - The system is busy
    - Please call later
  - Using Erlang C

$$P_W = \frac{\frac{\rho^c}{c!} \frac{c}{c-\rho}}{\sum_{i=0}^{c-1} \frac{\rho^i}{i!} + \frac{\rho^c}{c!} \frac{c}{c-\rho}} = \frac{P_B \frac{c}{c-\rho}}{1 + \frac{P_B \rho}{c-\rho}}$$

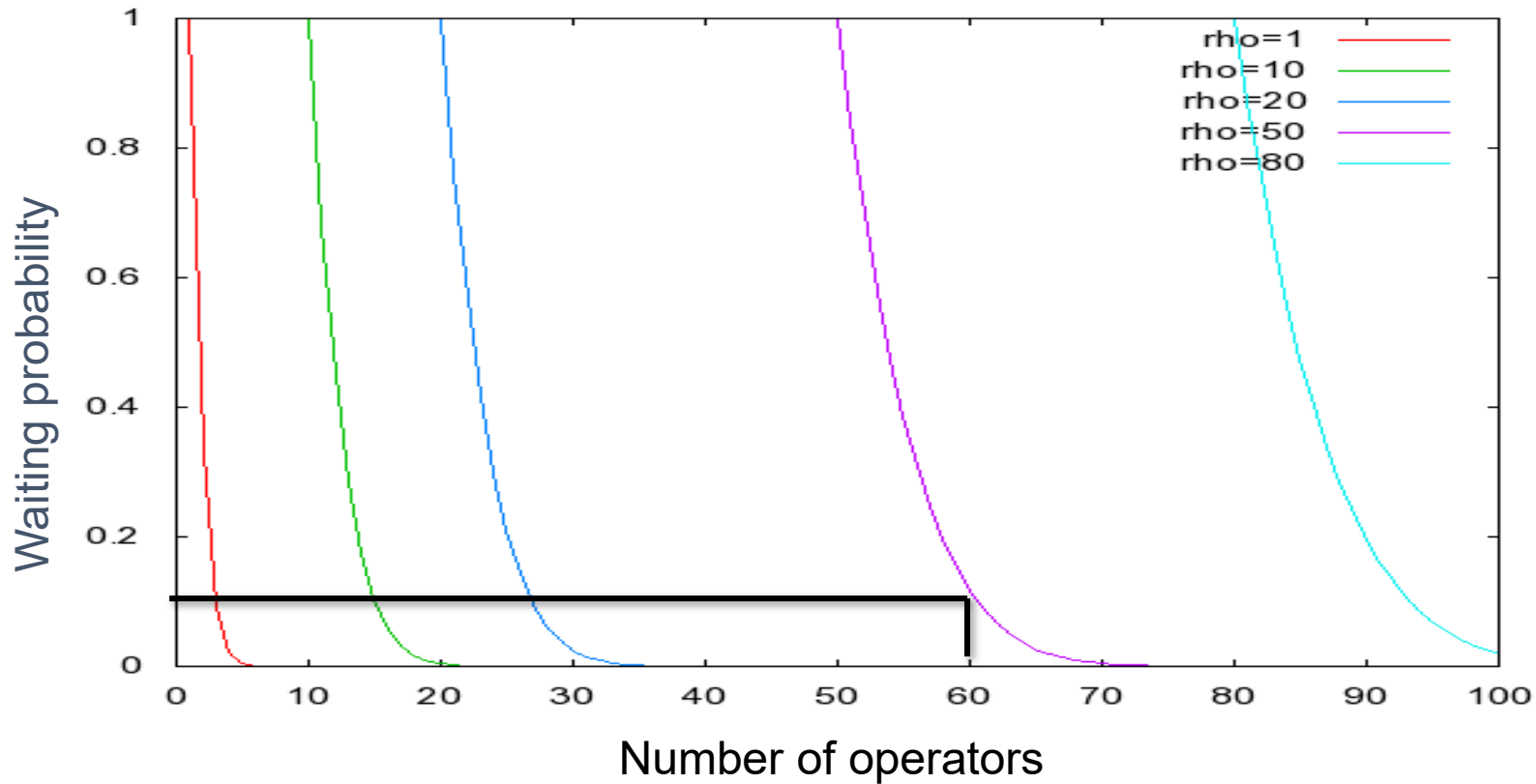


# Probability of waiting: 待つ確率



- Number of operators: 50
- On average: 40 calls come per minute, one call = 1 minute
- The waiting probability is 0.06 (6%)

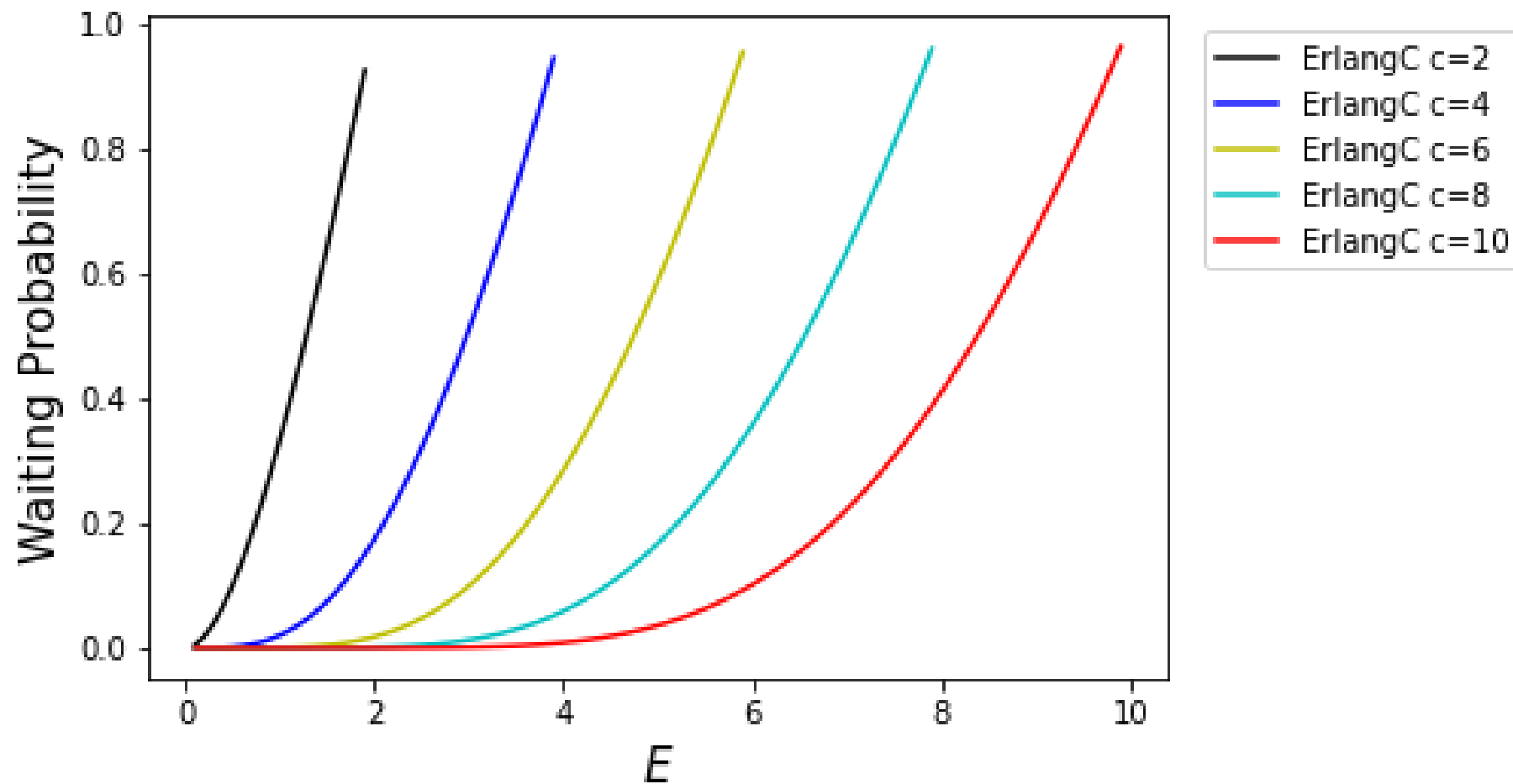
# Determine the number of operators オペレータの最適な数を求めよ！



- 50 calls come per minute
- Waiting probability is less than 10%
  - Need more than 60 operators

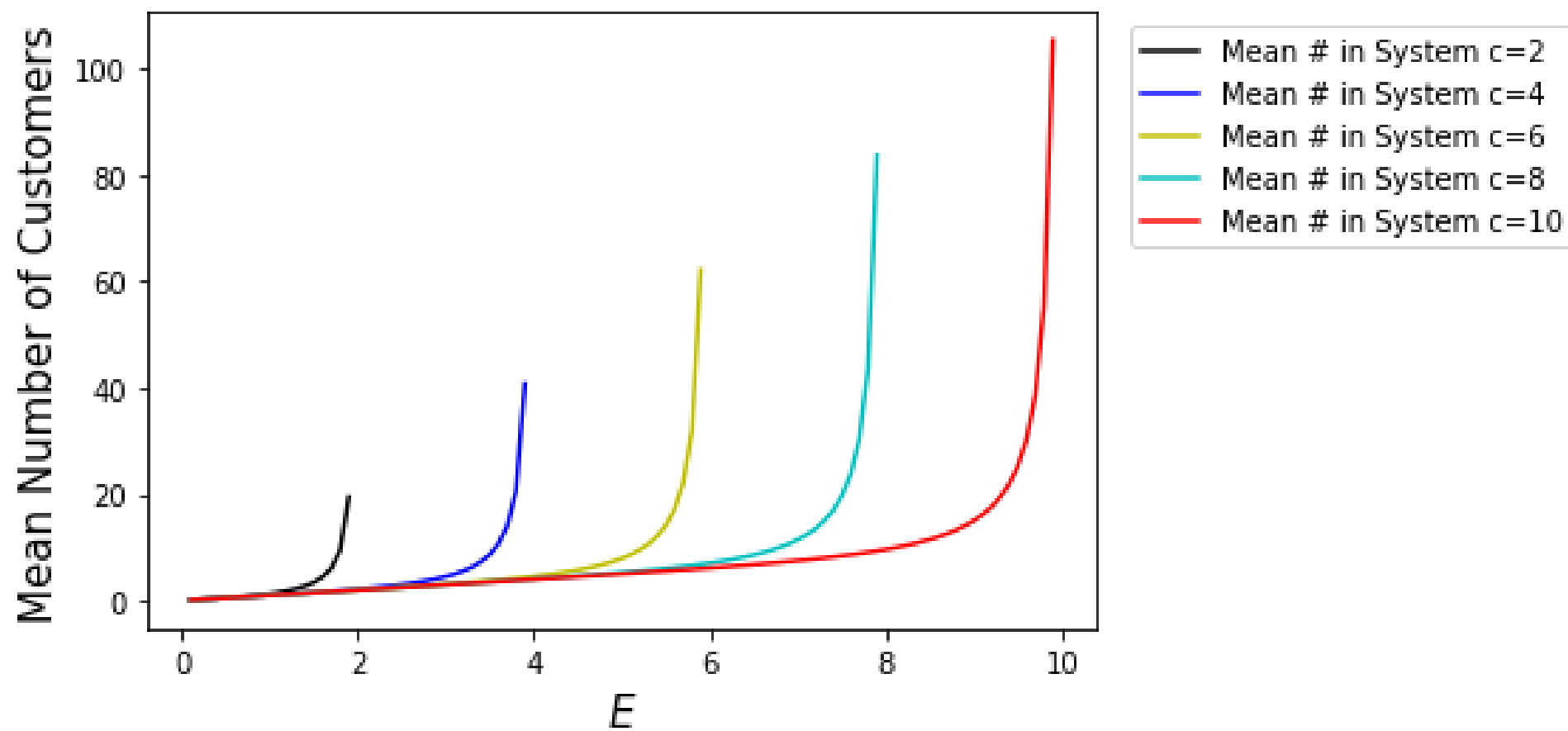
待つ確率

$$\mu = 1, E = \frac{\lambda}{\mu} = \lambda, \text{ 安定条件 } \lambda < c.$$

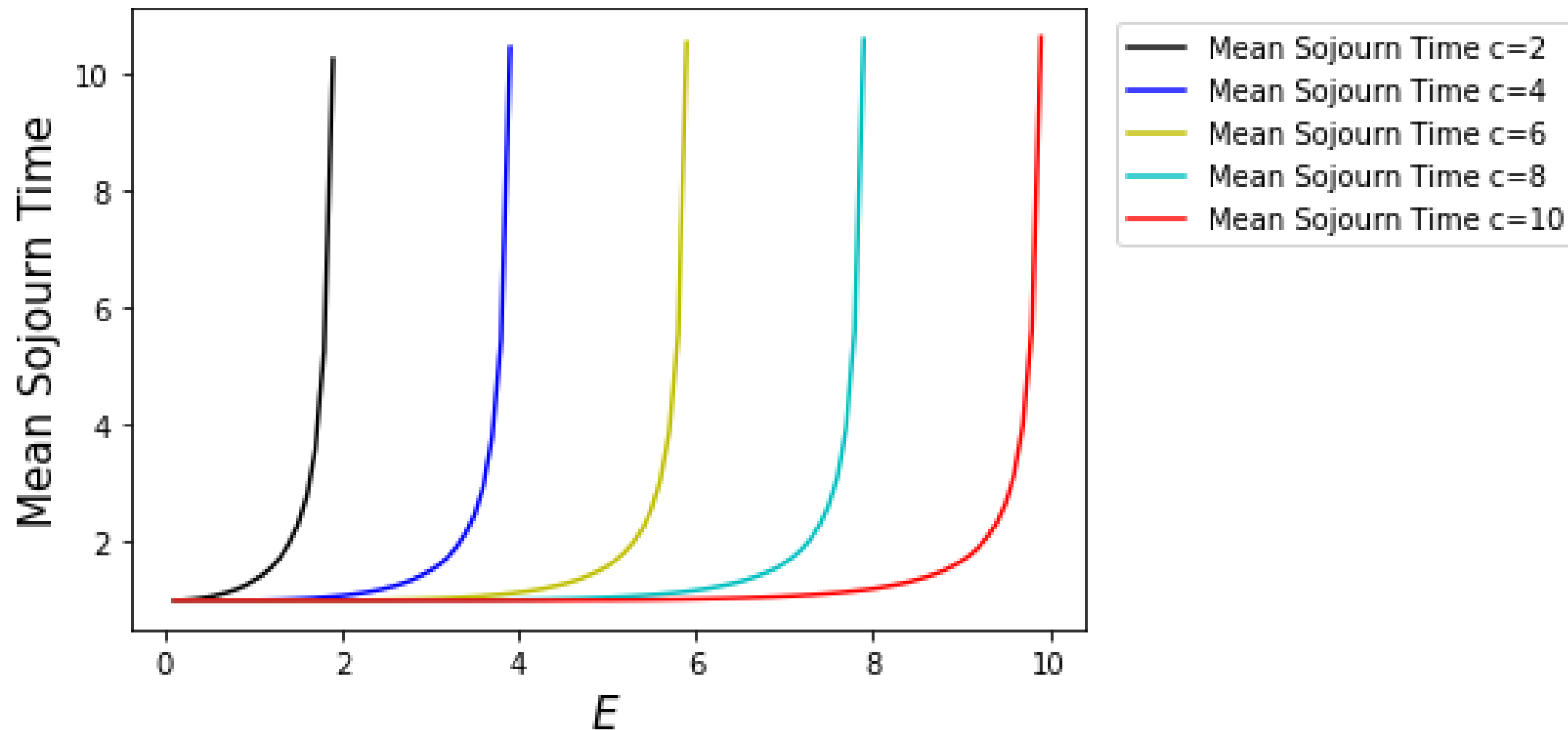




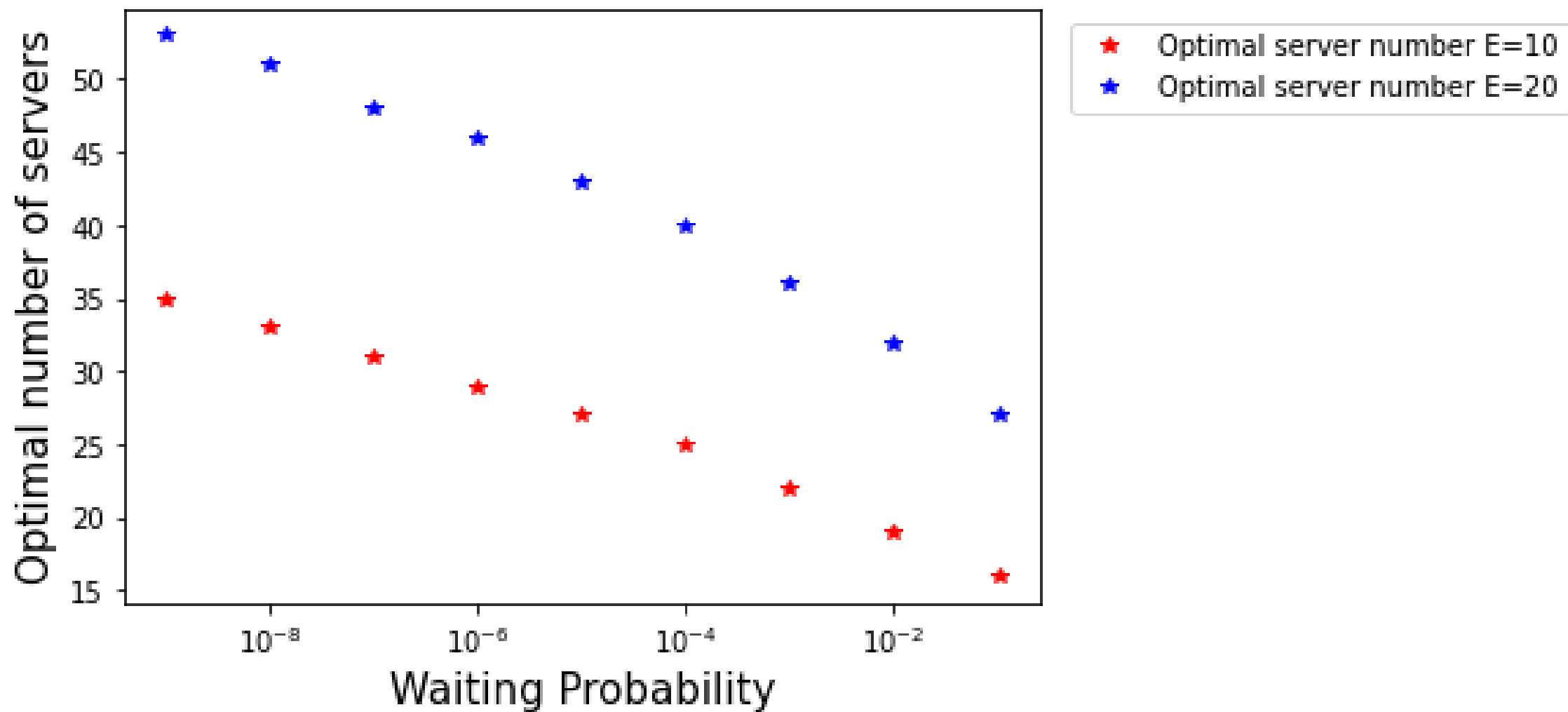
# 平均系内客数



平均滞在時間（入ってから出るまで）



# 最適なサーバ数



# レポート課題

- 課題 1

- ページ24のような設定で $c = 10$ の場合の待つ確率のグラフを作成せよ.
- また、待つ確率の曲線をシミュレーションと理論値の両方をプロットすること

- 課題2

- 課題1と同じ設定で縦軸をブロッキング確率にして、シミュレーションと理論のグラフを作成せよ.
- 課題 1 のグラフと比べて、ブロッキング確率と待つ確率の値を比較して、考察せよ.

- 課題3

- $M/M/c/c$ モデルと $M/M/c$ モデルにおいて、同じ到着率 $\lambda$ とサービス率 $\mu$ に対して、 $c$ を大きくしたときの系内客数の分布をプロットしてその変化を考察してください. また、その時、パラメータ $\rho = \lambda/\mu$ のPoisson分布と比較せよ.
- ただし、確率変数 $X$ がPoisson分布に従うとは $P(X = n) = e^{-\rho} \frac{\rho^n}{n!}, n = 0, 1, \dots$